WEEK 8 PROJECT REPORT

STUDENT PERFORMANCE DASHBOARD USING POWER BI

Submitted By: Divya B
Roll Nor: E25DA013
Department: Data Science and Artificial Intelligence
Tool Used: Microsoft Power BI
Project: Student Performance Dashboard – Academic Performance Analysis

1. INTRODUCTION

The Student Performance Dashboard is a data visualization project developed using Microsoft Power BI. It is designed to analyze student academic performance through charts, graphs, KPI cards, and interactive slicers.

The dashboard presents information about subject scores, pass and fail results, grade distribution, test preparation courses, parental level of education, and lunch categories. It also provides an Academic Profile page to view individual student details using drill-through functionality.

2. OBJECTIVES

- To analyze students' academic performance.
- To compare average scores across subjects.
- To visualize pass and fail distribution.
- To analyze student grade distribution.
- To examine performance across different categories.
- To create an interactive dashboard using Power BI.
- To display individual student details using drill-through.

3. TOOLS USED

Microsoft Power BI: Used to create the dashboard and visualizations.

Power Query Editor: Used to prepare and transform the dataset.

Charts and KPI Cards: Used to display academic information visually.

DAX: Used for calculated measures, where required.

4. DATASET DESCRIPTION

The dataset contains student information and academic scores. The important fields include:

- Student ID
- Gender
- Race/Ethnicity
- Parental Level of Education

- Lunch
- Test Preparation Course
- Math Score
- Reading Score
- Writing Score
- Subject and Score
- Grade and Pass/Fail Result

These fields help analyze student performance from different perspectives.

5. DATA PREPARATION

The dataset was imported into Power BI and prepared for analysis. The relevant columns were selected and organized for visualization.

The Math, Reading, and Writing score columns were transformed into Subject and Score columns using the unpivot operation, where applicable. The prepared data was then used to create charts, KPI cards, and slicers.

6. DASHBOARD DEVELOPMENT

6.1 Page 1 – Student Performance Dashboard

Step 1: Dashboard Title

Added the main title, *Student Performance Dashboard*, to identify the purpose of the report.

Step 2: KPI Cards

Created KPI cards to display important performance measures, including average scores and pass percentage.

Step 3: Average Score by Subject

Created a chart to compare average scores across different subjects.

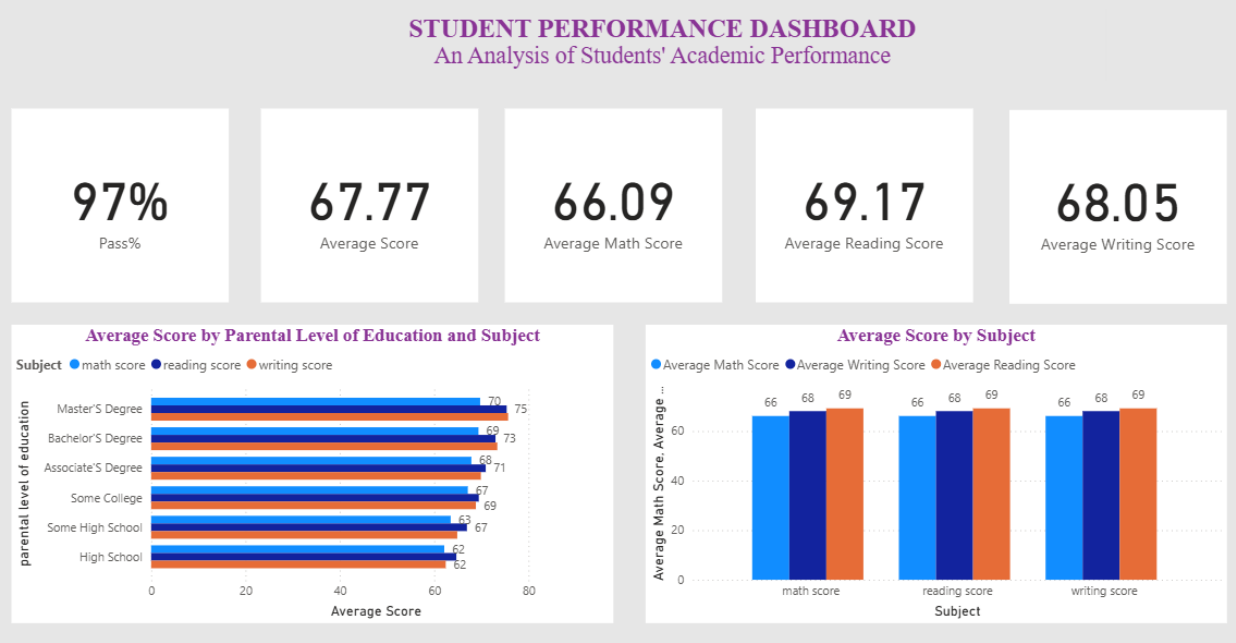
Step 4: Pass and Fail Distribution

Added a chart to display the distribution of students according to their pass and fail results.

Step 5: Test Preparation Course Distribution

Created a chart to show the distribution of students based on their test preparation course category.

Purpose: Page 1 provides an overall summary of student performance through KPI cards and visualizations.



6.2 Page 2 – Student Performance Analysis

Step 1: Page Title

Added a title to identify the page as a student performance analysis report.

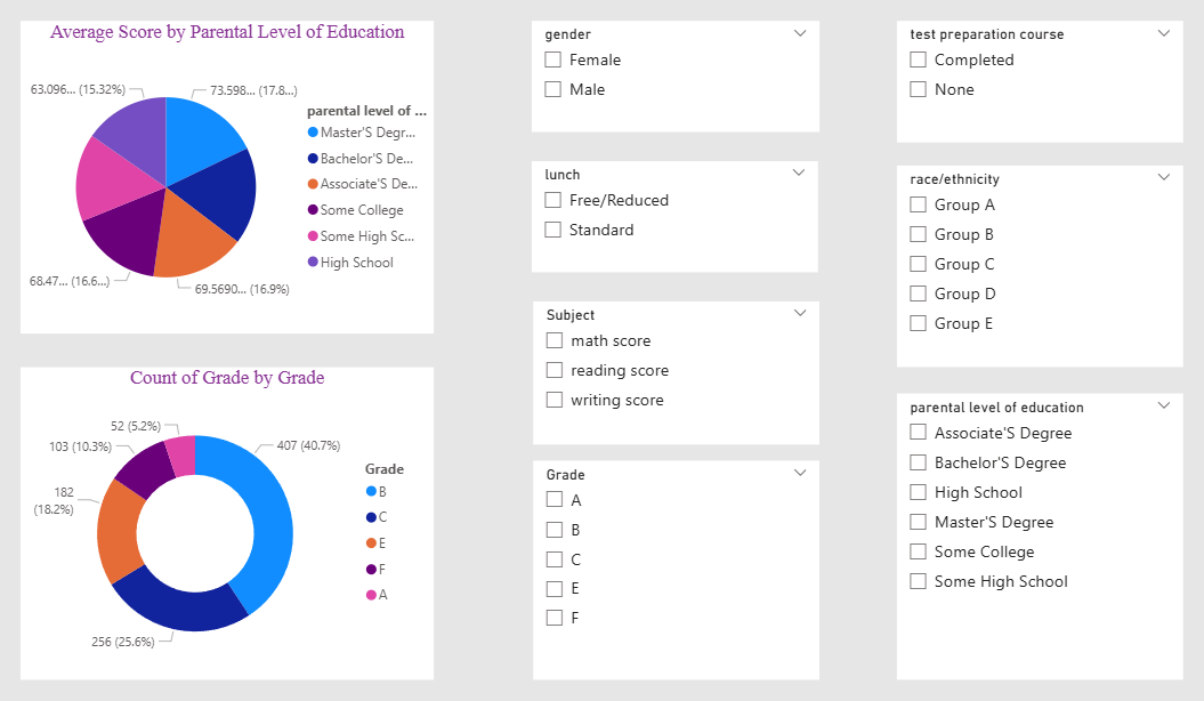
Step 2: Average Score by Parental Level of Education

Created a pie chart to compare average scores across parental education categories.

Step 3: Student Grade Distribution

Created a doughnut chart to display the distribution of students across grade categories.

Purpose: Page 2 helps users compare average scores by parental education and understand the distribution of student grades.



6.3 Page 3 – Detailed Performance Analysis

Step 1: Average Score by Test Preparation Course

Created a chart to compare average scores across test preparation categories.

Step 2: Average Score by Parental Level of Education

Added a chart to examine average score differences across parental education categories.

Step 3: Average Score by Lunch Category

Created a chart to compare average scores across the available lunch categories.

Step 4: Student ID Slicer

Added a Student ID slicer to allow users to select a student and filter the connected visualizations.

Purpose: Page 3 provides detailed comparisons of average scores and supports interactive student-level filtering.



6.4 Page 4 – Student Performance Analysis

Step 1: Performance Visualizations

Arranged the available charts to provide an additional view of student performance.

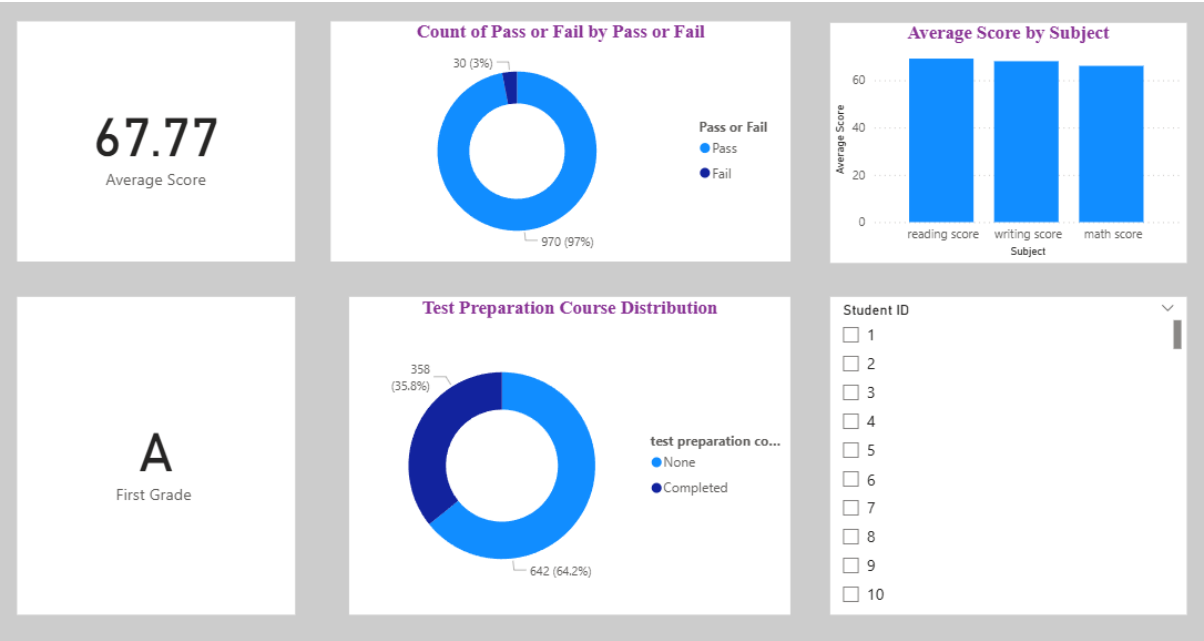
Step 2: Filtering Options

Used the available slicers to explore student records and categories.

Step 3: Visual Interaction

Checked the connected visuals to ensure that they respond to slicer selections as configured.

Purpose: Page 4 provides additional performance analysis and allows users to explore student data interactively.



6.5 Academic Profile – Drill-through Page

Step 1: Created the Academic Profile Page

Created a separate page to display detailed student information.

Step 2: Added Student ID to Drill-through

Added Student ID to the drill-through field area so the page can receive the selected student's filter context.

Step 3: Created the Student Details Table

Created a table containing available details such as Student ID, Gender, Grade, Average Score, Lunch, Parental Level of Education, Pass/Fail Result, Race/Ethnicity, Test Preparation Course, Subject, and Score.

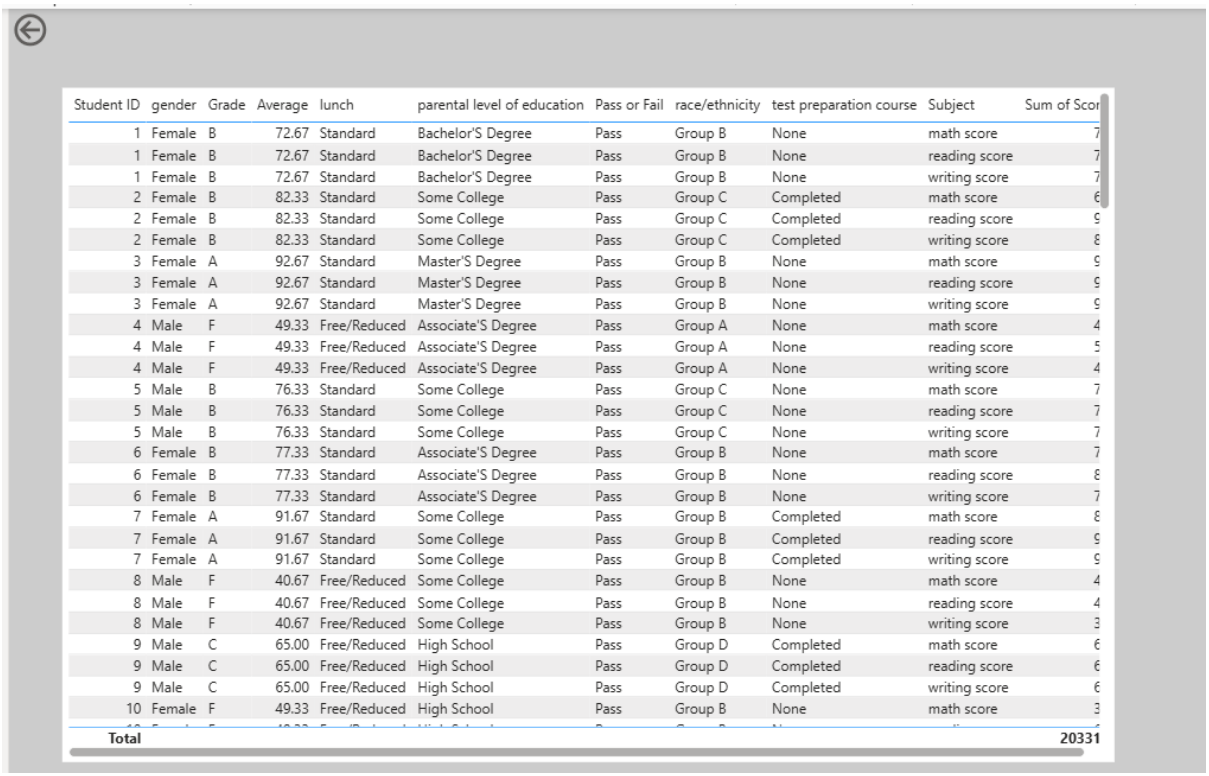
Step 4: Added the Back Button

Added a Back button to help users return to the previous report page.

Step 5: Tested Drill-through Functionality

Selected a student record and opened the Academic Profile page through drill-through to check whether the displayed information matched the selected student.

Purpose: The Academic Profile page allows users to view individual student details and navigate back to the main report.



Student ID	gender	Grade	Average	lunch	parental level of education	Pass or Fail	race/ethnicity	test preparation course	Subject	Sum of Scor
1	Female	B	72.67	Standard	Bachelor'S Degree	Pass	Group B	None	math score	7
1	Female	B	72.67	Standard	Bachelor'S Degree	Pass	Group B	None	reading score	7
1	Female	B	72.67	Standard	Bachelor'S Degree	Pass	Group B	None	writing score	7
2	Female	B	82.33	Standard	Some College	Pass	Group C	Completed	math score	6
2	Female	B	82.33	Standard	Some College	Pass	Group C	Completed	reading score	9
2	Female	B	82.33	Standard	Some College	Pass	Group C	Completed	writing score	8
3	Female	A	92.67	Standard	Master'S Degree	Pass	Group B	None	math score	9
3	Female	A	92.67	Standard	Master'S Degree	Pass	Group B	None	reading score	9
3	Female	A	92.67	Standard	Master'S Degree	Pass	Group B	None	writing score	9
4	Male	F	49.33	Free/Reduced	Associate'S Degree	Pass	Group A	None	math score	4
4	Male	F	49.33	Free/Reduced	Associate'S Degree	Pass	Group A	None	reading score	5
4	Male	F	49.33	Free/Reduced	Associate'S Degree	Pass	Group A	None	writing score	4
5	Male	B	76.33	Standard	Some College	Pass	Group C	None	math score	7
5	Male	B	76.33	Standard	Some College	Pass	Group C	None	reading score	7
5	Male	B	76.33	Standard	Some College	Pass	Group C	None	writing score	7
6	Female	B	77.33	Standard	Associate'S Degree	Pass	Group B	None	math score	7
6	Female	B	77.33	Standard	Associate'S Degree	Pass	Group B	None	reading score	8
6	Female	B	77.33	Standard	Associate'S Degree	Pass	Group B	None	writing score	7
7	Female	A	91.67	Standard	Some College	Pass	Group B	Completed	math score	8
7	Female	A	91.67	Standard	Some College	Pass	Group B	Completed	reading score	9
7	Female	A	91.67	Standard	Some College	Pass	Group B	Completed	writing score	9
8	Male	F	40.67	Free/Reduced	Some College	Pass	Group B	None	math score	4
8	Male	F	40.67	Free/Reduced	Some College	Pass	Group B	None	reading score	4
8	Male	F	40.67	Free/Reduced	Some College	Pass	Group B	None	writing score	3
9	Male	C	65.00	Free/Reduced	High School	Pass	Group D	Completed	math score	6
9	Male	C	65.00	Free/Reduced	High School	Pass	Group D	Completed	reading score	6
9	Male	C	65.00	Free/Reduced	High School	Pass	Group D	Completed	writing score	6
10	Female	F	49.33	Free/Reduced	High School	Pass	Group B	None	math score	3
Total										20331

7. INTERACTIVE FEATURES

The dashboard includes the following features:

- **KPI Cards:** Display important performance measures.
- **Charts:** Present student data visually.
- **Slicers:** Filter data based on selected categories.
- **Drill-through:** Display detailed information about a selected student.
- **Back Button:** Return to the previous report page.
- **Visual Interactions:** Allow connected charts to respond to selections when configured.

These features make the dashboard interactive and easy to explore.

8. KEY OBSERVATIONS

The dashboard helps users compare subject scores, examine pass and fail distribution, understand grade distribution, and explore average scores across different categories.

The Student ID slicer and Academic Profile page make it easier to examine individual student records. The visualizations provide a clear view of the available academic data.

09. CONCLUSION

The Student Performance Dashboard was developed using Microsoft Power BI to analyze and visualize student academic data. It includes KPI cards, charts, slicers, and drill-through functionality.

The project provides an organized way to explore student scores, grade distribution, and performance across different categories. Through this project, practical knowledge was gained in data preparation, data visualization, dashboard development, and interactive reporting.